

AVINASH KUMAR K S

QA Engineer — Python Test Automation

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PROFESSIONAL SUMMARY

Quality Assurance professional with 5+ years at Amazon, including 3.5+ years in Python-based test automation for Echo device platforms. Grew internally from Quality Associate to Testing Associate to Quality Assurance Technician. Proposed and built the OTA test automation initiative from proof-of-concept through full production rollout across 7 Echo device families, eliminating a manual process that took approximately 1 hour per test case per device. Later owned the Developer App Quality (DAQ) component and directed a team of 10 device associates. Strong in Python, Pytest, and device/firmware test automation. Building UI automation depth in Playwright.

TECHNICAL SKILLS

Test Automation: Python, Pytest, Playwright, REST API testing (Postman)

CI/CD & Tools: GitHub Actions, Git, JIRA, Confluence, TestRail, internal Amazon pipelines

Testing Practices: Functional, Regression, Integration, Smoke & Sanity, Exploratory, STLC, defect lifecycle, Agile/Scrum

Frameworks & Debugging: FastAPI, Pydantic, Streamlit, SQLite, Claude API, SQL, Bash, Linux shell (grep, tail, find, awk, ps), Kibana

AI-Augmented Engineering: LLM evaluation, prompt engineering, AI-assisted testing, agentic workflows, Claude API, Amazon Q

PROFESSIONAL EXPERIENCE

Quality Assurance Technician

Dec 2024 - Jan 2026

Amazon Development Center | Chennai, India

- Took full ownership of the OTA test automation initiative following promotion, building on the proof-of-concept from Testing Associate. Collaborated with QAs and SDETs to build out the automation library, write test cases, and deploy across 7 Echo device families in 2 weeks — eliminating a manual process that previously took approximately 1 hour per test case per device. Code reviewed and approved by Senior SDETs and QAEs; still running in production today.
- Triaged daily pipeline executions and automation runs throughout the year, fixing failures and growing automation coverage. Automated 180+ additional test cases (approximately 5 minutes each to run manually), removing an estimated 15+ hours of manual execution per regression cycle.
- Owned the Developer App Quality (DAQ) component for approximately 3–4 months — the pre-release quality gate for Echo device builds released to 3rd-party developers. Directed a team of 10 device associates (dotted-line reporting) on daily test execution, bug triage, and validation cycles. Release ownership was handled by a senior QAE based in the US.
- Proposed the concept for an AI-driven test failure analysis approach to help identify root causes faster; shared as a process-improvement idea within the team (not implemented at Amazon) — later built independently as failureIQ (see Projects).
- Resolved two hard-to-diagnose pipeline failures: traced a MediaTransform Player HTTP 403 error to a recent endpoint auth change affecting a second component's pipeline, and repaired a test that had been failing for ~1 month by identifying app-side log drift rather than marking it as expected failure.
- Built an OCR-based text-movement validation function (using Amazon Q for code generation with own design and integration) that filled a long-standing automation gap. Explicitly recognized by a Lead SDET.
- Onboarded two new React Native apps (Settings, Video Player) — built by SDETs to enable testing against APIs from other teams as device apps moved off C++ — validating functionality, authoring test plans and manual test cases for device associates to execute, then automating 30 of the 40 test cases.

Testing Associate

Jun 2022 - Nov 2024

Amazon Development Center | Chennai, India

- Proposed the OTA test automation concept and built the initial Proof-of-Concept after OTA testing had been fully manual since 2022 (6–8 cases × 7 devices daily, each taking approximately 1 hour to execute manually). Designed a Python/Pytest framework using AI-assisted development, automating 2 of 4 OTA steps via the device's internal CLI. The PoC directly contributed to promotion to Quality Assurance Technician and an SDM/QAM-sponsored rollout across all 7 device families.

- Executed daily functional, regression, and exploratory testing on Echo device builds. Logged defects with reproducible steps, screenshots, and device logs. Verified fixes through standard retest workflow.
- Built a Python tool for JIRA metrics extraction using JQL and the JIRA REST API, reducing manual reporting time from approximately 30–45 minutes to 5–10 minutes.

Quality Associate (Customer Support)

Aug 2020 – May 2022

Amazon Development Center | Chennai, India

- Joined Amazon directly out of college. Handled customer chats and calls with a focus on first-contact resolution.
- Transitioned internally to Testing Associate through Amazon's Internal Job Posting (IJP) program — selected for the technical track based on consistent performance.

PERSONAL PROJECTS

failureIQ — AI-Powered Test Failure Investigation Engine

Jul 2026 – Present

github.com/akaAvinash/failureIQ | Python, MIT license

- Built an AI-powered QA investigation engine that sits on top of any test automation framework and automatically root-causes failing tests, producing a shareable HTML report in under two minutes.
- Designed a framework-agnostic plugin architecture with three integrations — pytest/Playwright, Robot Framework (Selenium/Appium), and a generic JSON ingest path — using the Claude API to determine root cause, confidence score, and a suggested fix.

Interview Agent — Open Source AI-Powered Interview Prep Platform

Apr 2026 – Present

[interview-agent-av.streamlit.app](https://github.com/akaAvinash/interview-agent) | github.com/akaAvinash/interview-agent | Python, FastAPI, Pytest, Pydantic, Streamlit, SQLite, Claude API, Pyodide

- Designed and shipped an autonomous interview-prep agent that parses a resume and job description, generates a tailored multi-topic interview, conducts it as a chat, and scores answers using an LLM-as-judge to produce a feedback report.
- Architected a Protocol-based question source layer with four implementations and two deployment modes — hosted (free, bring-your-own-key, Pyodide browser execution) and self-hosted with a FastAPI backend.

Home Lab Agent — Self-Hosted AI Monitoring & Auto-Remediation

Aug 2026

Python, Docker, systemd, Qwen3-8B (local LLM via llama-CLI), Claude API

- Built a self-hosted AI monitoring and auto-remediation agent for a home Linux server, combining a local LLM with Claude API escalation to diagnose system, Docker, and systemd issues.
- Designed a safety-first architecture where AI output is always re-validated against a code-enforced whitelist before any action runs; shipped with dry-run mode and SQLite-backed incident tracking.

TP-Link VPN Automation Tool

Aug 2026

Python, Playwright

- Built a Python automation tool using Playwright with layered retry logic to make VPN client connection handling more reliable.
- Tuned settle delays to match actual UI state lag, reducing unnecessary wait time in the automation flow.

EDUCATION

Bachelor of Technology in Mechanical Engineering

2015 – 2019

Kerala Technological University, India

CERTIFICATIONS & RECOGNITION

- Software Engineer Certificate — HackerRank
- Amazon: Ace of Innovation Award · Quarterback Champion (Q3 2023) · Top Ideator, Ideathon (Q4 2023, 24 valid ideas) · RnR Award

LANGUAGES

English, Hindi, Malayalam, Tamil